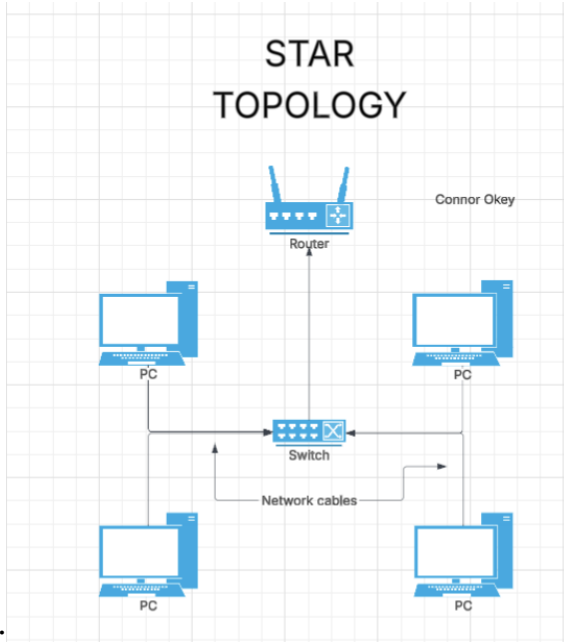
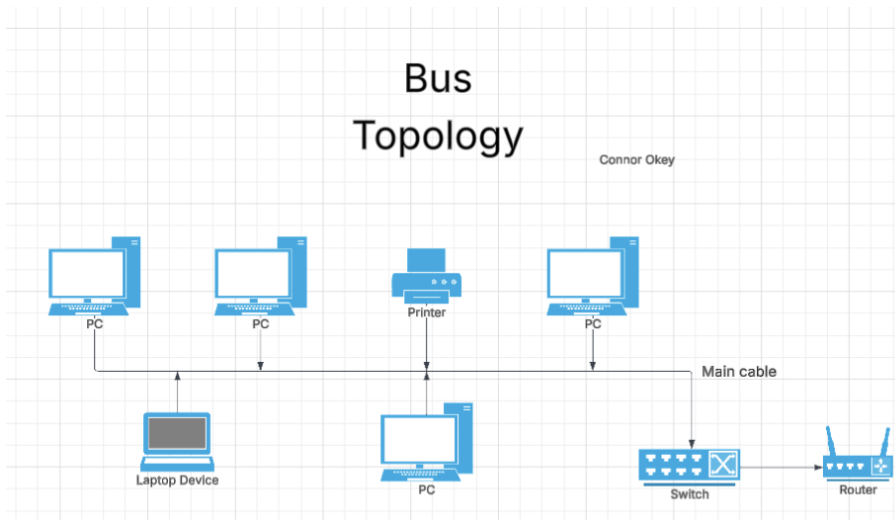
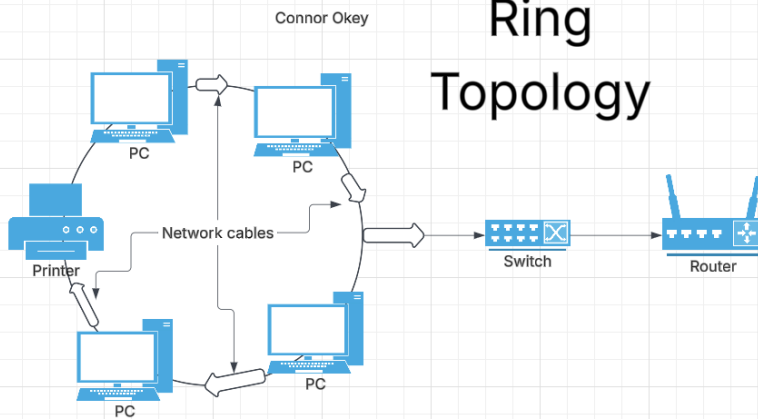




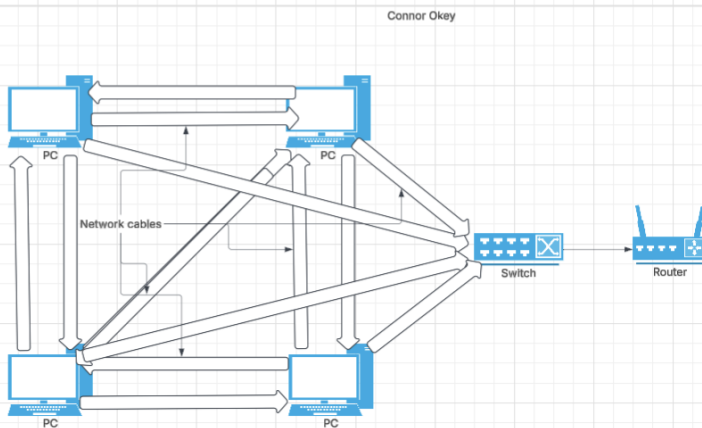
PART 1:



Ring Topology



MESH TOPOLOGY



PART 2:

A company uses a star topology. All computers connect to a central switch. Suddenly, all devices lose network access at the same time.

- What is the MOST likely cause?

The most likely cause of the loss of network would be the switch failing.

- Why would this issue affect all devices?

All the devices are connected to the main switch, they send information to the switch and receive data. If the main switch goes out, then so does access to the network for all the devices connected.

- What are 2 possible fixes?

The first possible fix would be to restart the system by unplugging the central switch and plugging it back in. The next fix would be to get a new switch for the system.

- What is one improvement to prevent this in the future?

One improvement would be to have another switch that all the devices connect to. This way if one of the switches fail, the other should still work long enough to replace the broken one.

A company uses bus topology. Users report very slow network speeds, especially during peak hours.

- What is causing the slowdown?

Multiple computers sending and receiving data through one main cable. This naturally causes a bottleneck because one cable can only handle so much data being moved at once.

- Why is this common in this topology?

When so much data from different devices is being transferred the network capacity decreases as more devices try to use the network.

- What is one better topology to replace it with?

The star topology.

- Why is that better?

Each computer having its own cable to connect to the switch eliminates the single point of failure. If a cable fails, only the one device loses connection and the network stays online. It also creates paths to each device from the switch so there won't be data collisions between devices.

1. Which topology is best for a data center with high east-west traffic?

The spine and leaf topology works the best for high east-west traffic.

2. How is a client-server network different from a peer-to-peer network?

Client server networks rely on servers to manage data and requests, which means if the server goes out then so does anyone's access to the data in the server until it goes back up. Peer to peer networks allow devices to communicate without a server which eliminates a single point of failure. While client server networks are more expensive, they are secure and more complicated they are far more secure compared to peer to peer networks.

3. What are the three layers in the hierarchical model?

The first layer is the access layer. This is what allows the authorized to access the network with ease. The next layer is the distribution layer which connects the access layer to the rest of the world. The last layer is the core layer. The core layer is what is responsible for moving large amounts of data between a data center and the main employee locations.

4. What type of network connects personal devices over short distances?

The personal area network.

5. What is the main advantage of a full-mesh WAN topology?

It's highly fault tolerant because some links can be lost but reachability between all sites can still be maintained.

6. Write a reflection on your thoughts of this assignment, what was hard easy etc.

The hardest part of this assignment was learning about the different Topologies and having to make a diagram of them. Since the textbook didn't cover some of them, I had to look them up to learn about the different topologies. Once I figured out the design of each of the topologies was literally in the name it made it way easier to remember what they were and their pros and cons.

The easiest part of the assignment was once I learned the designs were literally in the name writing those up was easy for me to do. I appreciate that I had a hard time figuring out the different topologies, that has helped me to better understand and remember them.